

Social equity indicators for a Blue Economy: Guidelines for application

Cisneros-Montemayor, Breaky, Bossier, Croft, Issifu, Keefer, Singh, Voyer, Ota (2025). *Marine Policy*.

This paper explores the types of indicators we can use to guide the Blue Economy: total output indicators (how much), disaggregated indicators (how much among who and where), and equity action indicators (what is being done for who and where). They note how the ‘Blue Economy’ movement was originally intended to focus on aspects of social equity but is now used in a wide variety of contexts that can be confounded with ‘Blue Growth’.

Introduction

“Social equity goals are specifically stated in most new ocean management and development plans, notably those aligned with a Blue Economy that recognizes the significant and rising inequities across ocean spaces and sectors. However, the deeper significance of social equity has still not been fully integrated in policy and management practices, which often ask why it is relevant and how it can be measured in ways comparable to more familiar ecological and economic indicators.”

“Acceleration in ocean industries such as mariculture, bioprospecting, and renewable energy [12], [41], [57]—has rapidly increased new economic opportunities but done so in ways that benefit only a few. This does not only exclude people from benefiting from new industries, but often actively dispossesses and marginalizes local communities from access to their traditional territories and resources [10], [12], [21], [44]. A focus on sustainable economic growth is therefore not enough to deliver truly equitable development, ...”

“This paper builds on research into multidisciplinary indicators, which has highlighted a particular need for ways track progress on equity and justice goals in marine sectors [18], [45], [54], but has not yet explicitly connected recommendations from social justice scholarship and civil rights movements to help choose appropriate indicators.”

“... we propose using established principles addressing class, race, gender and colonial injustice to guide the design of Blue Economy indicators.”

“...while some general indicators can be useful across cases, successfully planning and implementing local blue economies requires a critical examination of their specific context, which necessarily involves the feedback and leadership of frontline communities—that is, those directly involved in and impacted by the Blue Economy.”

“... it is important to emphasize that a focus on social equity was indeed intended to set the Blue Economy apart from pure economic quantification of the ocean economy, or more narrow growth-oriented industrialization goals of ‘blue growth’ that undermine environmental sustainability [20]. However, the term ‘Blue Economy’ is now used by many different actors and under a correspondingly wide set of emphases [61], [76], and is even viewed skeptically by some marginalized groups that the proponents of its original meaning intended to support [15].”

“[We] undertake a more in-depth discussion of the equity and justice principles that underpin the consideration and selection of equity-focused indicators... Finally, we discuss benefits and examples that transition from absolute outcomes to disaggregated outcomes to process indicators.”

“Croft et al. [27] proposed a model for incorporating social equity within blue economies by defining goals and objectives that are centered on four main principles drawn from key research in the field (see, e.g., [6], [28], [43])—Achieving Fairness, Preventing Harm, Respecting Rights, and Supporting Flourishing.”

Key Principles for Indicators in an Equitable Blue Economy

“Indicators are often proxies for concepts that are multidimensional, context-specific, and difficult to measure directly. As highlighted above, social equity is a particularly complex goal but can be defined in practice [27,28].”

“To aid in the selection of indicators for an equitable Blue Economy, from total outputs to disaggregated impacts to equity actions (discussed at length in Section 4), in this section we highlight four principles (Difference, Intersectionality, Power Relations, and Interest Convergence) based on fundamental and cross-cutting aspects of social (in)justice and subordination, such as class (economic), gender, colonization and race.”

• Difference

“The Difference principle holds that equal liberties and opportunities must be attained, and equal opportunities to contribute to decisions be granted to ensure equitable decision-making and outcomes.”

“Any treatment of equity has to take into account historic injustices that shaped contemporary circumstances and acknowledge the accumulated advantage that some groups have gained over others due to systemic biases and overt oppression. In practice this means a focus on disaggregated data which identifies those who face the most inequity procedurally and in outcomes.”

• Intersectionality

“How to identify those “left behind” is not trivial. We therefore suggest the second guiding principle to follow the principle of intersectionality [11], [26]. Intersectionality holds that people hold multiple identities and applying policies that differentiate people too coarsely will fail to appreciate the kinds of systematic pressures that different groups face. ... [We] suggest that any options for disaggregated indicators are considered through a range of possible dimensions of inequity (e.g., gender, race, age, ethnicity, nationality, etc.; see Table 1).”

• Power Relations

“... we suggest that any selection process of indicators must include representation from the groups that equity-serving interventions are supposed to serve. Neglecting to include these groups in a co-produced process of prioritization risks selecting indicators that do not reflect the priorities of the groups whom interventions are meant to benefit.”

• Interest Convergence

“Interest convergence recognizes that gains for marginalized groups usually only progress when they align with the interests of groups with the power to affect policy and lead interventions...”

“These effects are clear across ocean spaces, where (neo)colonial decision-making did not consider local benefits unless they were beneficial for the metropole, with negative effects on development and capacity persisting for centuries afterward [23].”

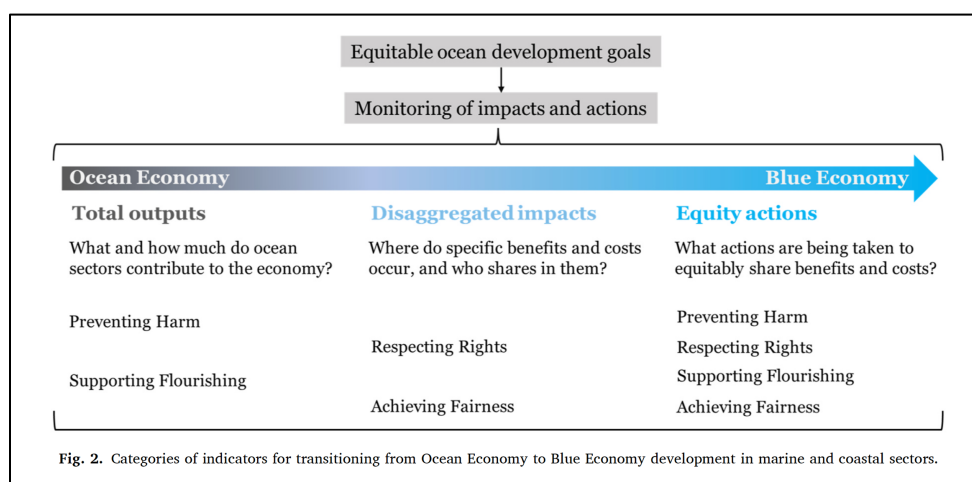
“... to guard against the risks that interest convergence presents, we suggest operationalizing this principle by limiting the role of external proponents of interventions in selecting and monitoring indicators (unless the proponent is a currently excluded community).”

Indicators for an Equitable Blue Economy

“Whether using absolute or relative indicators, tracking policy progress in general can follow one of four approaches based on comparing current states to 1) a threshold (whether an ideal or current standard); 2) past conditions; 3) a proxy (assumed to be directly correlated with the ultimate goal); or 4) a counterfactual condition (i.e., expected outcomes without an intervention). All four kinds of measurement are useful and are often done together, but can also be misleading and misused.”

“Comparisons to a threshold can be meaningful where there are specific measurable objectives, such as drinking water quality (e.g., microbial loads). People without access to drinking water is a clear sign of inequity. However, only defining success against an ideal standard may detract from observable progress towards that standard.”

“Indicators can be divided into three main categories as one transitions from pure quantification of an ocean economy to a transformational approach under a Blue Economy: 1) Total outputs, 2) Disaggregated impacts, and 3) Equity actions (Fig. 2).”



• Total Output Indicators

“While total output indicators are generally quite coarse, the general lack of data on economic dynamics in many ocean sectors means that these indicators may sometimes be the only form of standardized data collected in a country or region. That being the case, they have historically been used as implied proxies for social well-being, for example by assuming that there are population-wide gains (or losses) in seafood supply or income when ocean sectors grow. Indeed, these total output indicators can be complemented with local-based understandings of social processes and events affecting ocean sectors to provide deeper insights into community well-being and can sometimes flag emerging issues. Clearly, however, this is insufficient to adequately gauge the effects of new policies and their strategies, particularly when the distribution of impacts is a primary focus.”

• **Disaggregated impact indicators**

“Disaggregated impact indicators specifically distinguish who shares in benefits and/or bears costs from ocean sectors (Table 1). This also considers where benefits are produced (e.g., where a fish was caught) compared to where they are enjoyed (e.g., where it is eaten, or where value is added). In the context of equitable development, this is an absolutely vital improvement over total output indicators.”

• **Equity Action Indicators**

“Equity action indicators track what is being done to support equitable processes and outcomes, and to combat existing systemic or individual inequities; these are anti-inequity and anti-racist actions. In order to truly advance towards an equitable Blue Economy, indicators related to equity actions, such as anti-racism and decolonization, are an essential next step to disaggregated indicators because they focus on the specific initiatives being taken to advance equity in a specific context.”

Table 1

Worked examples of indicators for tracking progress on social equity goals in a Blue Economy, with descriptions of indicator types including Total Outputs, Disaggregated Impacts, and Equity actions.

Indicator Category	Indicator examples
Total Outputs <i>How much?</i> Total contribution of ocean sectors to economies at particular scales. This can also include effects of ocean sectors on ecosystems (these are impacts on natural capital).	<i>Contributions to well-being</i> ■ Fisheries catch, landed value, stock status. ■ Aquaculture production, trade value. ■ Fishworkers employed (full or part-time). ■ Tourist expenditures and employment generated by marine ecotourism. ■ Revenue received from blue carbon credits. ■ Energy generated offshore. <i>Impacts on ecosystem services and natural capital</i> ■ Number of overfished stocks. ■ Water pollution from coastal activity and/or agriculture. ■ Restoration of habitat quality for nutrient cycling. ■ Repopulation of local species. ■ Re/De-forestation of coastal vegetation. ■ Space used by offshore facilities.
Disaggregated Impacts <i>Who and where?</i> Similar metrics to total output indicators, but specifically collected for different groups of people.	<i>Contributions and Impacts by groups based on (e.g.):</i> ■ Ethnicity ■ Gender ■ Race ■ Geographic area ■ Indigeneity ■ Internal/Foreign Migrants ■ Language ■ Modes of practice (e.g., artisanal, industrial) ■ Nationality ■ Political affiliation ■ Wealth/Class ■ Traditional ownership
Equity Actions <i>What is being done?</i> Initiatives and actions taken to improve equity outcomes and processes.	<i>Following from identified groups above:</i> ■ Financial support, by group. ■ Conferral of ownership in sectors, by group. ■ Delegation of management decisions, by group. ■ Participation in earliest stages of planning, by group. ■ Hiring of planners and managers, by group.

Conclusion

“The guidelines and examples of indicators presented here will hopefully help inform planning and decisions for more equitable Blue Economies across the world, but these decisions will always have to take local contexts into account. At the very least, we urge readers and practitioners to

thoughtfully consider who should benefit from a Blue Economy, how they can be freed from any subordinated relationships, and how these processes and their expected outcomes can be measured most directly to ensure we are doing all we can do to protect the interest of frontline and marginalized populations across the world's oceans.”